



GREEN UNIVERSITY OF BANGLADESH

Department of Computer Science and Engineering



Course Outline

1 General Information

Spring 2025 CSE 313	Faculty	Faculty of Science and Engineering (FSE)
	Department	Department of Computer Science and Engineering (CSE)
	Programme	Bachelor of Science in Computer Science and Engineering
	Semester	Spring 2025
	Course Title	<i>Software Engineering</i>
	Course Code	CSE 313
	Course Credit	3.0 units
	Contact Hours	2.5/week
	Course Status	Core Course
	Prerequisite Course	None

2 Course Instructors

Section	Name	Office	Email
222D2	Sheikh Fazle Rabbi	A608	rabbi@cse.green.edu.bd
223D2	Fatema Tuj Johora	A510	fatema@cse.green.edu.bd
223D3	Md. Riad Hassan	A510	riad@cse.green.edu.bd
223E2 231E2	Md. Riad Hassan	A510	riad@cse.green.edu.bd

3 Class Hours

Section	Room	Weekday	Time	Weekday	Time
222D2	K106	Wednesday	10:30 AM - 11:45 AM	Friday	10:30 AM - 11:45 AM
223D2	J109 J107	Sunday	01:30 PM - 03:00 PM	Tuesday	11:30 AM - 01:00 PM
223D3	K 106	Monday	10.20 AM - 11.40 AM	Wednesday	10.20 AM - 11.40 AM
223E2 231E2	Online	Wednesday	06:30 PM - 09:00 PM		

4 Counseling Hours

Section	Weekday	Time	Weekday	Time
222D2	Friday	11:00 AM - 12:15 PM	Thursday	08:30 PM - 09:45 PM
223D2	Tuesday	09.00 AM - 10.10 AM		
223D3	Monday	11.40 AM - 01:00 PM		
223E2 231E2	Tuesday	6:30 PM - 9:00 PM		

5 Course Rationale

To provide a comprehensive introduction to software engineering leading to the ability to understand contemporary terminology, progress, issues, and trends. A thorough introduction to software engineering techniques, focusing on software process models, agile development. Principle that guides practice, understanding requirements, Requirements modeling: scenarios, information and analysis classes, Requirements modeling: Flow, behavior, patterns and web apps, Design concept, pattern-based design, quality management, software quality assurance, software testing strategies.

6 Course Objectives

This course aims to achieve the following objectives:

- Acquiring a comprehensive understanding of the foundational principles and processes in software engineering.
- Developing the ability to apply modern software engineering techniques to analyze, design, implement, and maintain software solutions.
- Acquiring the skills to evaluate and maintain software systems for continuous improvement.

7 Course Description

Concepts/paradigms/phases of Software Engineering/life cycle; Recurring and Fundamental Principles; Requirement Analysis/Engineering/Specification; Concepts of Feasibility Analysis and techniques of cost benefit analysis; Role, Tasks and Attributes of System Analysts ; Software Architectures; Design: Architectural Design, Module Design, Interfaces and fundamental Principles; User Interface (UI) Design: UI Design Principles, User Interaction, Information Presentation, User Support, Interface Evaluation; Implementation Issues; Introduction to Software Testing; Quality Assurance; Configuration Management: Version Management and Tools; Maintenance and Evolution; Project Management: Software process and project Metrics, Software Project Planning, Risk Analysis and management, Project Scheduling and Tracking; Use case model for requirement writing; Elaboration using System Sequence Diagram; UML diagrams; Interaction and Collaboration Diagram for designing Software; GRASP patterns.

8 Teaching Methods

Maximum topics will be covered from the textbook. For the rest of the topics, reference books will be followed. Some class notes will be uploaded on the web. Multimedia projector will be used most of the time for the convenience of the students. For some cases, white boards will be used for beefing in detail. Students must participate in classroom discussions for case studies, problems solving and project developments.

9 Course Outcomes

CO	CO Description	PO	Domain (LoBT)	Weight	WK	WP	EA	Assessment Methods
CO1	Illustrate the core concepts of software engineering for designing the solutions of real-life engineering problems.	PO1	Cognitive (C3)	40%				

CO2	Analyze real-life problem domains to make effective solutions using software engineering development knowledge, including requirement discovering process, proper SDLC model selection, and system modeling.	PO2	Cognitive (C4)	50%	Please refer to SECTION 10 .
CO3	Design software systems, taking industrial strength into consideration, for sustainable development by identifying the impacts of societal and environmental contexts.	PO7	Affective (A4)	10%	

Legend:**CO:** Course Outcome**PO:** Program Outcome (**APPENDIX: A**)**WK:** Knowledge Profile (**APPENDIX: B**)**WP:** Complex Problem Solving (**APPENDIX: C**)**EA:** Complex Engineering Activities (**APPENDIX: D**)**LoBT:** Level of Bloom's Taxonomy (**APPENDIX: E**)**10 Assessment Methods of COs**

Assessment Method	CO1	CO2	CO3	Total
Final Exam	20%	20%		40%
Midterm Exam	20%	10%		30%
Class Tests	5%	5%		10%
Presentation(KSA 01)		5%	5%	10%
Quiz, Attendance(KSA 02)		5%	5%	10%
Total	40%	50%	10%	100%

11 Topic Outline

Lecture	Selected Topic	Article	Problems
(1-2)	Professional software development	T 1.1	1.1, 1.2,1.3, 1.5, 1.8
	Software engineering ethics	T 1.2	
(3-5)	Software process models	T 2.1	2.1, 2.3, 2.4, 2.6, 2.7, 2.9
	Process activities	T 2.2	
	Coping with change	T 2.3	
	The rational unified process	T 2.4	
(6-9)	Agile methods	T 3.1	3.1, 3.2, 3.4, 3.5,3.6, 3.9
	Plan-driven and agile development	T 3.2	
	Extreme programming	T 3.3	
	Agile project management	T 3.4	
	Scaling agile methods	T 3.5	
(10-12)	Functional and non-functional requirements	T 4.1	4.2,4.3,4.4, 4.6, 4.7
	The software requirements document	T 4.2	
	Requirements specification	T 4.3	
	Requirements engineering processes	T 4.4	
	Requirements elicitation and analysis	T 4.5	

	Requirements validation	T 4.6	
	Requirements management	T 4.7	
(13-15)	Context models	T 5.1	
	Interaction models	T 5.2	
	Structural models	T 5.3	
	Behavioral models	T 5.4	
	Model-driven engineering	T 5.6	
(16-18)	Architectural design decisions	T 6.1	
	Architectural views	T 6.2	6.1,6.2,6.3,
	Architectural patterns	T 6.3	6.7, 6.8
	Application architectures	T 6.4	
(19-20)	Object-oriented design using the UML	T 7.1	
	Design patterns	T 7.2	
	Implementation issues	T 7.3	
	Open source development	T 7.4	
(21-22)	Development testing	T 8.1	
	Test-driven development	T 8.2	
	Release testing	T 8.3	10.1,10.4,10.5,
	User testing	T 8.4	10.7,10.10
	Evolution processes	T 9.1	
	Software maintenance	T 9.3	
(23-26)	Dependability properties	T 11.1	
	Availability and reliability	T 11.2	11.1, 11.3,
	Safety and Security	T 11.3.	11.5
		11.4	
(27-30)	Risk management	T 22.1	
	Managing people	T 22.2	
	Teamwork	T 22.3	
	Software pricing	T 23.1	
	Project scheduling	T 23.3	22.1, 22.3, 23.2,
	Change management	T 25.1	25.5, 26.2, 26.4
	Version management	T 25.2	
	The process improvement process	T 26.1	
	Process change	T 26.4	

For the definitions of T and R, Please refer to Section 12.

12 Text and Reference Materials

T Textbook:

- Ian Sommerville, **Software Engineering**, 10th Edition, Pearson, 2015.

R References:

- Roger Pressman, & Bruce Maxim, **Software Engineering: A Practitioner's Approach**, 9th edition, McGraw-Hill Education, 2019.

13 Grading Policy

Marks Obtained	Letter Grade	Numerical Evaluation	Definition
80% and above	A+	4.00	Excellent
75% <80%	A	3.75	Excellent

70% < 75%	A-	3.50	Very Good
65% < 70%	B+	3.25	Good
60% < 65%	B	3.00	Good
55% < 60%	B-	2.75	Good
50% < 55%	C+	2.50	Average
45% < 50%	C	2.25	Average
40% < 45%	D	2.00	Below Average
below 40%	F	0.00	Failing

14 Additional Course Policies

1. **Equipment and Aids:** Bring your own materials such as a calculator, notebook, and pen to participate effectively in classroom activities. You are NOT allowed to borrow from others inside the classroom which may potentially create distractions for your classmates.
2. **Assignments:** There will be a number of assignments for formative assessment purposes. The average of the assignment marks will be used for computing the final grade. Late submission of homework will carry a zero mark.
3. **Class Tests:** There will be at least three Class Tests taken during the semester and the best two will be counted for final grading. A class test can be taken with/without prior announcement.
4. **Examinations:** The midterm and final examinations will be a closed book, closed notes. Mobile phones are strictly prohibited in the exam hall. Please bring your own watch (non-smart) and synchronize at the beginning of the examination.
5. **Test Policy:** In case of missing a test without prior notice to the respected faculty member, a zero mark will be given. No makeup tests will be taken as the best two test scores will be considered for grading out of three tests.
6. **Mobile Devices Policy:** Empirical evidence of using multitasking devices such as laptops and smartphones in the classroom hinders the learning experience. Thus, the use of multitasking devices is strictly discouraged. Switch off your laptop/mobile devices during class activities.

15 Additional Information

Please click or scan:

ACADEMIC CALENDAR SPRING, 2025:



ACADEMIC INFORMATION AND POLICIES:



PROCTORIAL RULES:



GRADING AND PERFORMANCE EVALUATION:



Sheikh Fazle Rabbi
Course Coordinator, CSE 313
February 01, 2025

Dr. Muhammad Aminur Rahaman
Chairman, Department of CSE
February 01, 2025

Appendix A : Program Outcomes

POs	Category	Program Outcomes
PO1	Engineering Knowledge	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis	Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.
PO3	Design/Development of Solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.
PO4	Investigations	Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
PO5	Modern tool usage	Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	Environment and sustainability	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics	Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice.
PO9	Individual work and teamwork	Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.
PO10	Communication	Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects in multidisciplinary environments.
PO12	Life Long Learning	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

Appendix B : Knowledge Profile

Knowledge Profile	Attribute
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline
WK2	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge that supports engineering design in a practice area
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability

Appendix C : Range of Complex Engineering Problem Solving

Attribute	Identity	Complex Engineering Problem Description
Depth of knowledge required	WP1	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
Range of conflicting requirements	WP2	Involve wide-ranging or conflicting technical, engineering and other issues
Depth of analysis required	WP3	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
Familiarity of issues	WP4	Involve infrequently encountered issues
Extent of applicable codes	WP5	Are outside problems encompassed by standards and codes of practice for professional engineering
Extent of stakeholder involvement and conflicting requirements	WP6	Involve diverse groups of stakeholders with widely varying needs
Interdependence	WP7	Are high-level problems including many component parts or sub-problems

Note: Complex Engineering Problems have **IDENTITY P1 AND SOME OR ALL OF P2 TO P7**.

Appendix D : Range of Complex Engineering Activities

Attribute	Identity	Activity Description
Range of resources	EA1	Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies)
Level of interaction	EA2	Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues
Innovation	EA3	Involve creative use of engineering principles and researchbased knowledge in novel ways
Consequences for society and the environment	EA4	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	EA5	Can extend beyond previous experiences by applying principles-based approaches

Note: Complex activities means (engineering) activities or projects that have **SOME OR ALL OF THE ABOVE ACTIVITIES**.

Appendix E : Domain and Level of Bloom's Taxonomy

Cognitive Domain		Psychomotor Domain		Affective Domain	
C1	Remembering	P1	Perception	A1	Receive
C2	Understanding	P2	Set	A2	Respond
C3	Applying	P3	Guided Response	A3	Value
C4	Analyzing	P4	Mechanism	A4	Organize
C5	Evaluating	P5	Complex Overt Response	A5	Internalize
C6	Creating/ Designing	P6	Adaption		
		P7	Origination		